Getting the redis-leader-deployment pod:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl apply -f https://k8s.io/examples/application/guestbook/redis-leader-
deployment.yaml
deployment.apps/redis-leader created
```

Searching for the pod:

```
NAME                              READY   STATUS    RESTARTS          AGE
backend-66569f457c-h9g2j          1/1     Running   5 (2m12s ago)     3d3h
backend-66569f457c-k94hh          1/1     Running   5 (2m12s ago)     3d3h
backend-66569f457c-mpdpv          1/1     Running   5 (2m12s ago)     3d3h
frontend-6fcd66b77d-qr8f5         1/1     Running   16 (117s ago)     3d3h
redis-leader-6465bf85d4-p29lm     1/1     Running   0                 37s
secret-configmap-pod              1/1     Running   2 (2m12s ago)     4h13m
```

The deployment logs:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl logs -f deployment/redis-leader

                Redis 2.8.19 (00000000/0) 64 bit

                Running in stand alone mode
                Port: 6379
                PID: 1

                      http://redis.io

[1] 04 May 13:44:12.546 # Server started, Redis version 2.8.19
[1] 04 May 13:44:12.547 # WARNING you have Transparent Huge Pages (THP) support enabled in your kernel. This wil
l create latency and memory usage issues with Redis. To fix this issue run the command 'echo never > /sys/kernel
/mm/transparent_hugepage/enabled' as root, and add it to your /etc/rc.local in order to retain the setting after
 a reboot. Redis must be restarted after THP is disabled.
[1] 04 May 13:44:12.547 * The server is now ready to accept connections on port 6379
```

Getting the redis-service pod

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ $ kubectl apply -https://k8s.io/examples/application/guestbook/redis-leader-
service.yaml
service/redis-leader created
```

Searching for the pod with 2 replicas:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl get pods
NAME                              READY   STATUS    RESTARTS          AGE
backend-66569f457c-h9g2j          1/1     Running   5 (6m4s ago)      3d3h
backend-66569f457c-k94hh          1/1     Running   5 (6m4s ago)      3d3h
backend-66569f457c-mpdpv          1/1     Running   5 (6m4s ago)      3d3h
frontend-6fcd66b77d-qr8f5         1/1     Running   16 (5m49s ago)    3d3h
redis-follower-6958c9f49c-7rcqv   1/1     Running   0                 28s
redis-follower-6958c9f49c-j57xp   1/1     Running   0                 28s
redis-leader-6465bf85d4-p29lm     1/1     Running   0                 4m29s
secret-configmap-pod              1/1     Running   2 (6m4s ago)      4h17m
```

Getting the redis-followers service:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl apply -f https://k8s.io/examples/application/guestbook/redis-followe
r-service.yaml
service/redis-follower created
```

Searching for the service:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl get svc
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
backend-service    ClusterIP   10.106.148.23    <none>        3000/TCP       3d3h
frontend-service   NodePort    10.99.175.111    <none>        80:32130/TCP   3d3h
kubernetes         ClusterIP   10.96.0.1        <none>        443/TCP        3d5h
redis-follower     ClusterIP   10.100.242.213   <none>        6379/TCP       80s
redis-leader       ClusterIP   10.99.252.145    <none>        6379/TCP       3m53s
```

Getting the frontend deployment:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl apply -f https://k8s.io/examples/application/guestbook/frontend-depl
oyment.yaml
deployment.apps/frontend created
```

Searching for the pod with the 3 replicas:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl get pods -l app=guestbook -l tier=frontend
NAME                        READY   STATUS    RESTARTS   AGE
frontend-88dcd974c-85pfr    1/1     Running   0          62s
frontend-88dcd974c-rqrtl    1/1     Running   0          62s
frontend-88dcd974c-z89z7    1/1     Running   0          62s
```

Getting the frontend service:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl apply -f https://k8s.io/examples/application/guestbook/frontend-serv
ice.yaml
service/frontend created
```

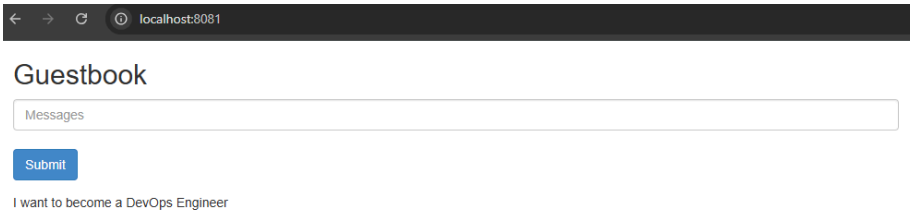
Searching for the service:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl get svc
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
backend-service    ClusterIP   10.106.148.23    <none>        3000/TCP       3d3h
frontend           ClusterIP   10.97.25.68      <none>        80/TCP         25s
frontend-service   NodePort    10.99.175.111    <none>        80:32130/TCP   3d3h
kubernetes         ClusterIP   10.96.0.1        <none>        443/TCP        3d6h
redis-follower     ClusterIP   10.100.242.213   <none>        6379/TCP       5m28s
redis-leader       ClusterIP   10.99.252.145    <none>        6379/TCP       8m1s
```

Running the PHP Guestbook browser:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl port-forward svc/frontend 8081:80
Forwarding from 127.0.0.1:8081 -> 80
Forwarding from [::1]:8081 -> 80
Handling connection for 8081
Handling connection for 8081
Handling connection for 8081
```

Inside the browser:



Now with the loadbalancer type on my frontend-service:

```
^Cjoaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl apply -f frontend-service.yaml
service/frontend configured
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl get svc frontend
NAME       TYPE           CLUSTER-IP    EXTERNAL-IP   PORT(S)        AGE
frontend   LoadBalancer   10.97.25.68   <pending>     80:30257/TCP   4m1s
```

Scaling up with 5 new replicas:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl scale deployment frontend --replicas=5
deployment.apps/frontend scaled
```

The logs of my 5 replicas:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl get pods -l app=guestbook -l tier=frontend
NAME                        READY   STATUS    RESTARTS   AGE
frontend-88dcd974c-85pfr    1/1     Running   0          7m49s
frontend-88dcd974c-9sz2v    1/1     Running   0          71s
frontend-88dcd974c-rqrtl    1/1     Running   0          7m49s
frontend-88dcd974c-ssgdl    1/1     Running   0          71s
frontend-88dcd974c-z89z7    1/1     Running   0          7m49s
```

Scaling out again for 2 replicas:

```
joaoribeiro@JOCAPC:~/PHP_GuestBook$ kubectl scale deployment frontend --replicas=2
deployment.apps/frontend scaled
```